

SMART INDIA HACKATHON 2026

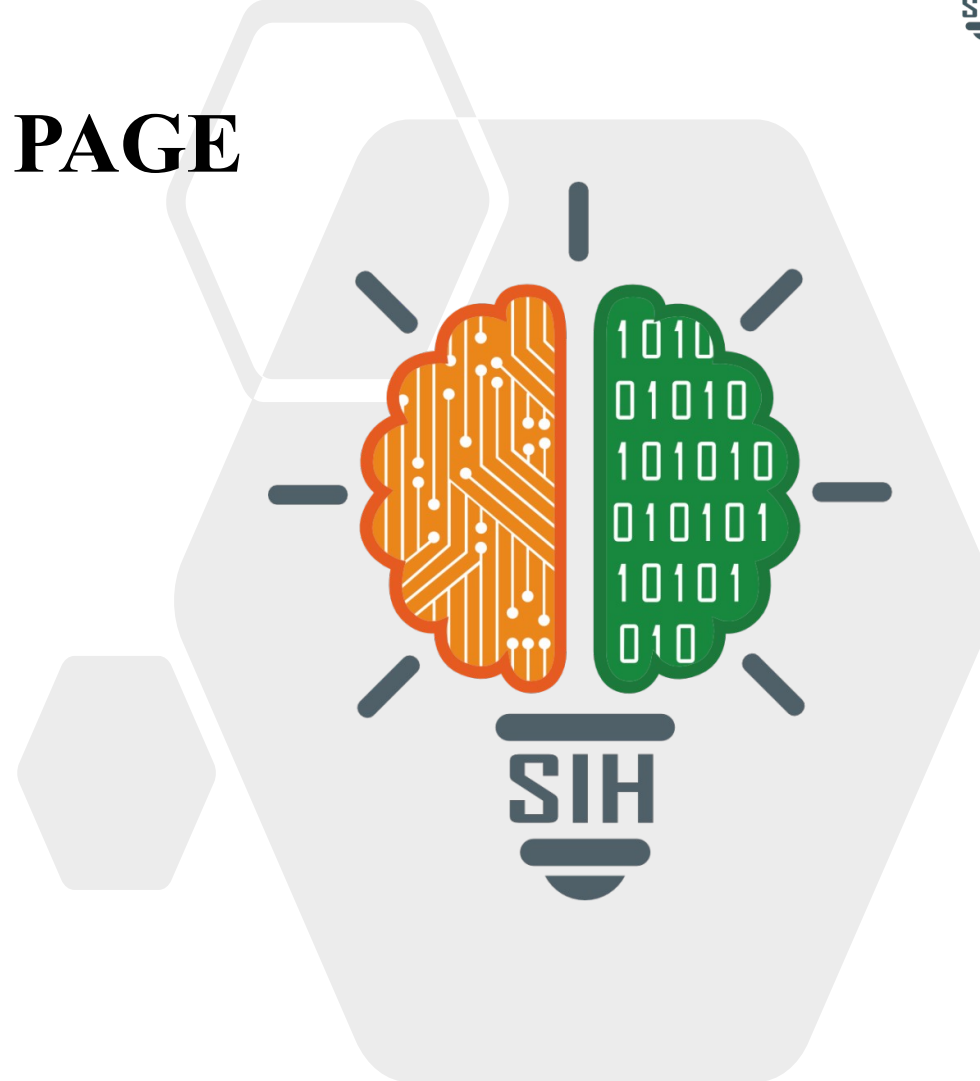


pramaan

Proof from any DVR.

TITLE PAGE

- **Problem Statement ID – 26150**
- **Problem Statement Title-** Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- <TEAM_ID>**
- **Team Name (Registered on portal) <TEAM_NAME>**



Pramaan — one workflow from any DVR disk to a Section 63-ready evidence package

Any DVR/NVR disk in; a signed, independently verifiable evidence package out. Fully offline.

❖ Proposed Solution — what it does

1 Image

read-only E01 / raw



2 Hash

SHA-256 + MD5



3 Identify OEM

→ tier A / B / C



4 Parse + recover

incl. deleted footage



5 Normalise time

four clocks → IST



6 Sign

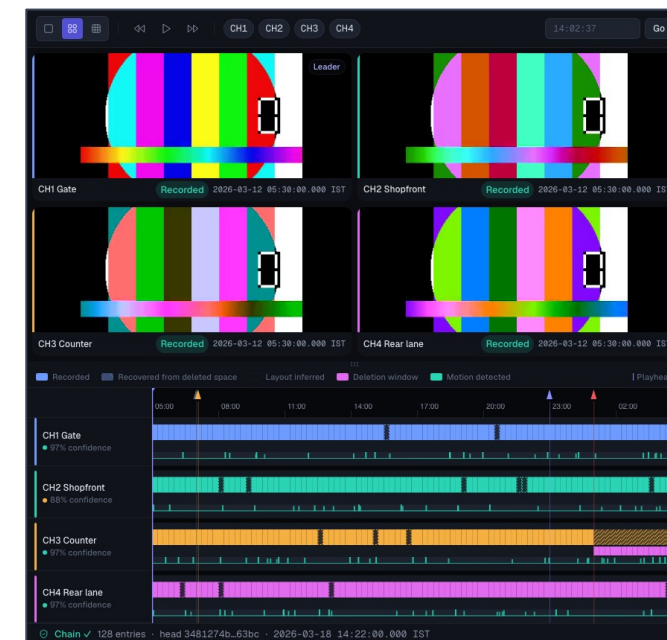
report + s.63 cert

❖ How it addresses the problem

- **Vendor-agnostic tiers:** A parse · B infer layout · C carve
- **No vendor player:** footage from every OEM in one workspace
- **Offline:** air-gapped workstation, no telemetry
- **Evidence read-only:** originals never re-encoded (stream copy)
- **Traceable:** every finding links back to disk bytes

❖ Innovation and uniqueness

- **Format inference:** undocumented vendors, proven on an unknown synthetic vendor
- **Four-clock time:** header, index, on-screen OCR, log clock changes
- **Deletion verdict:** method + actor from device logs
- **Custody:** Ed25519 hash chain + Merkle-anchored case root
- **Prove it:** any frame → its disk sectors + live SHA-256



Review workspace (synthetic case)

A deterministic Rust + Python forensic core; the LLM stays outside the evidence path.

❖ Technologies to be used

Rust scanner PyO3

Python 3.11 core

Kaitai .ksy specs

FFmpeg stream copy

DuckDB + Parquet

FastAPI + workers

React + TypeScript

PaddleOCR / ONNX

Ed25519 + Merkle

Claude API drafts only

```

01 10 Prove it
frm_fb181e680d08a685efb77985 20,992 bytes from offset 0x659d8

Verified — recomputed hash matches the stored claim
sha256 d5ca3d07..f59a stored d5ca3d07..f59a

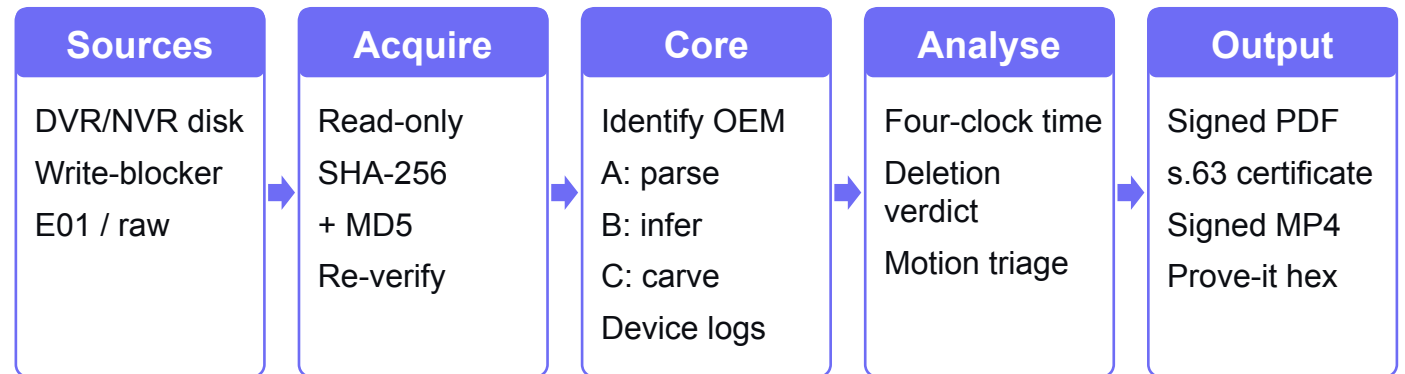
Before 256B < > After 512B < >

Offset      Hex      ASCII
000659d8 sec 812 2a 00 00 00 44 48 41 56 fc 00 03 00 90 00 00 00 *---DHAV-----
000659e8      2a 00 00 00 0b a0 d8 68 08 02 00 d1 00 00 00 01 *-----h-----
000659f8      41 9a 63 c0 1c cc 64 68 61 76 2a 00 00 00 44 48 A-c---dhav*---DH
00065a08 sec 813 41 56 fc 00 03 00 91 00 00 00 2a 00 00 00 0b a0 AV-----*-----
00065a18      d8 68 58 02 00 22 00 00 00 01 41 9a 83 c0 1c cc *hX---*---A-----
00065a28      64 68 61 76 2a 00 00 00 44 48 41 56 fc 00 03 00 dhav*---DHAV----
00065a38      92 00 00 00 2a 00 00 00 0b a0 d8 68 a8 02 00 73 *---*---h---s
00065a48      00 00 00 01 41 9a a3 c0 1c cc 64 68 61 76 2a 00 *---A---dhav*---
00065a58      00 00 44 48 41 56 fc 00 03 00 00 00 00 00 2a 00 DHAV-----*
  
```

Prove it: a recovered frame's exact disk bytes, rehashed live

❖ Methodology and process for implementation

Evidence path: one-way, read-only, deterministic



Custody, alongside every hop: Ed25519-signed, hash-chained audit log → case Merkle root anchor (local ledger today; Fabric next)

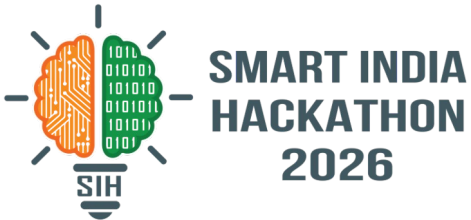
- LLM never sees frames or disk bytes
- Same input + version → identical outputs
- Every artefact records its provenance

Claude API (optional, off by default)

Structured metadata only; output is a labelled draft, never a finding



FEASIBILITY AND VIABILITY



Working prototype today: all 10 validation targets met on our synthetic corpus.

❖ Validation vs PRD targets (synthetic)

Metric	Target	Result	Scored
Deleted-frame recovery	≥ 95%	100%	5/10
Recovery precision	≥ 99%	100%	7/10
Recording parse	100%	100%	7/10
Timestamp error ≤ 1 s	≥ 99%	100%	9/10
Deletion method	100%	100%	6/10
Actor attribution	100%	100%	3/10
Tier B inference fields	100%	100%	2/10
Motion F1	≥ 80%	100%	6/10
Report determinism	yes	yes	10/10
Read-only guarantee	yes	yes	10/10

Synthetic corpus: 10 images modelled on published layouts (just validate). Real vendor disks not yet tested.

❖ Analysis of the feasibility of the idea

✓ Tier A parsers for 3 synthetic families

✓ Deleted footage recovered as playable, stream-copied clips

✓ Unknown-vendor layout inferred blind, then examiner-confirmed

✓ Signed PDF + BSA s.63 certificate, deterministic manifest

✓ Ed25519 custody chain + Merkle anchor, verified in UI

✓ E2E green at last run: 7/7 API, 7/7 browser

❖ Potential challenges and risks

Encrypted disks on new STQC-era firmware

Real disks differ from published layouts

Inferred layout mislabels a field

Poor on-screen clock OCR

LLM invents a claim in a draft

❖ Strategies for overcoming these challenges

Logical acquisition fallback (ONVIF / RTSP); limitation stated in the report

Real-disk corpus next: 5–6 used recorders, same validation harness

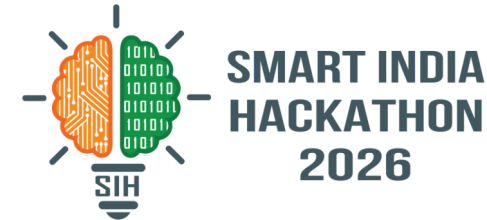
Confidence scores, examiner confirmation, tier labels A · B · C

Four clocks cross-checked; per-channel offset with confidence

Evidence-ID validator rejects unbacked sentences; draft label



IMPACT AND BENEFITS



One offline, court-ready workflow instead of vendor players and foreign suites.

❖ Potential impact on the target audience

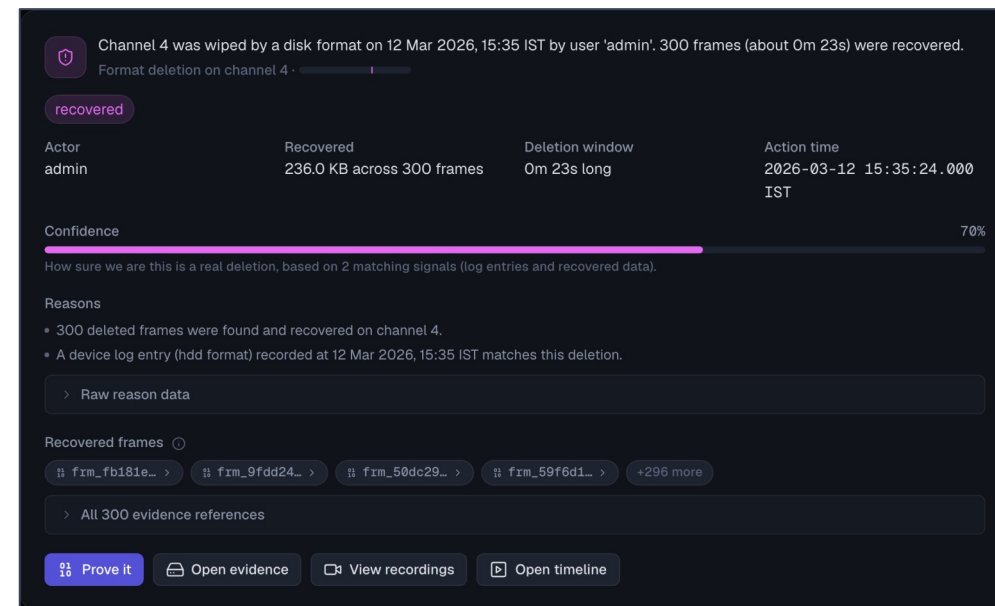
Forensic labs
one SOP

Cyber cells
fast triage

Courts
verifiable hashes

NTRC
unknown OEMs

- **Single tool:** one workflow in place of vendor players and foreign suites
- **Legal-ready:** auto-filled BSA s.63 certificate; fits BNSS s.176(3) video workflows
- **Recovery:** deleted footage, with method and actor from device logs
- **Triage:** synced multi-camera timeline, jump straight to incident time



Deletion verdict from the device log (synthetic case)

❖ Benefits of the solution

Social

- Faster justice: footage without vendor dependency
- Reproducible method that courts can verify

Economic

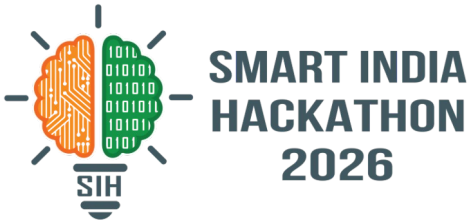
- No per-seat foreign licences; Make-in-India stack
- Open, reviewable format specs per vendor

Security

- Air-gapped: evidence never leaves the lab
- LLM off by default, metadata only



RESEARCH AND REFERENCES



Grounded in published DVR forensics research, open standards and Indian evidence law.

❖ Details / Links of the reference and research work

1	Han, Jeong & Lee · ICDF2C · 2015 Hikvision DVR file system → HIKSIM parser model	6	ONVIF Export File Format · IEC 62676-2-32 Signed export design (not yet conformance-tested)
2	Dragonas et al. · J. Forensic Sci. · 2023 / 2024 Hikvision and Dahua log records → actor attribution	7	NIST · NISTIR 8161 Rev. 1 CCTV digital video export profile → export design
3	Rzayeva et al. · Information · 2025 DVR recovery benchmark → our recovery target	8	BSA 2023 s.63 · Pune Bar Assn. v. UoI (2026) Certificate format → auto-filled s.63 template
4	Yoon & Hwang · DFRWS USA · 2026 Honeywell DVR file system → HWSIM parser model	9	NIST CFReDS · HeimVision DVR image Public DVR image → planned smoke test
5	SWGDE · DVR acquisition best practices · 2025 Seizure-clock SOP → intake fields, clock offset	10	MeitY / STQC · CCTV Essential Requirements Firmware security shift → encrypted-disk risk



Source code
github.com/MuaazSM/pramaan

Demo video
<DEMO_VIDEO_URL>

*All results so far are on our
synthetic corpus; real-disk
validation is the next milestone.*